

BENCHMARK: INACCURATE

Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2458016.8542 +/- 0.0017 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.169 +/- 0.051                 |
| Transit depth (Rp/Rs)^2                           | 2.84 +/- 1.73 [%]               |
| Orbital Inclination (inc)                         | 86.0 +/- 1.69                   |
| Airmass coefficient 1 (a1)                        | 5.66 +/- 3.55                   |
| Airmass coefficient 2 (a2)                        | -1.54 +/- 1.31                  |
| Scatter in the residuals of the lightcurve fit is | 67.72 %                         |
| Best Comparison Star                              | None                            |
| Optimal Aperture                                  | 2.88                            |
| Optimal Annulus                                   | 3.25                            |
| Transit Duration (day)                            | 0.039 +/- 0.022                 |

Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                       |
|--------------------------------------|-------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.169 $\pm$ 0.051 vs 0.131 (deviation 0.75 $\sigma$ ) |
| Duration fit vs published            | 0.039 d vs 0.1075 d (deviation 3.11 $\sigma$ )        |
| Mid-time O-C                         | -3.6 min                                              |
| Depth SNR                            | 3.3                                                   |
| Residual scatter                     | 67.72 %                                               |

Light curve

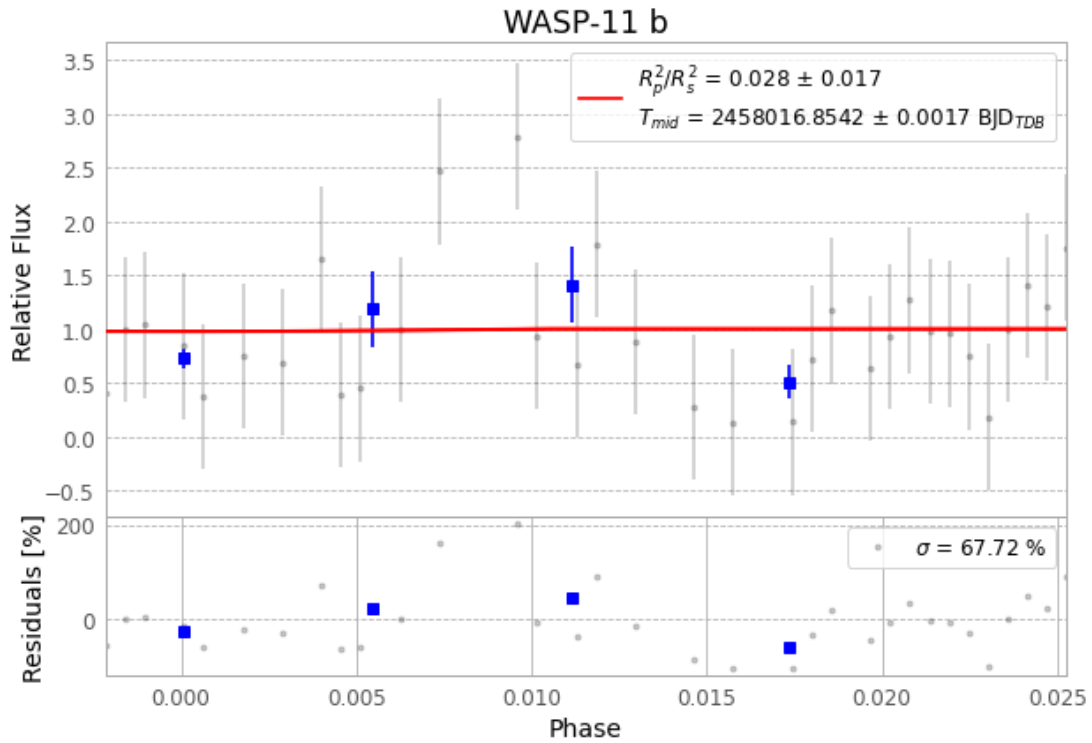


Figure 1 — FinalLightCurve\_WASP-11 b\_2017-09-20.png (SHA-256 019f33b9e4d073b0...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [545, 437], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 52a10dd371c0690a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ead60b9

## Data provenance

46 FITS frames (30 MB), HATP-10170920081212.FITS → HATP-10170920103912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-170920.log (f257b8bf1827...), FinalLightCurve\_WASP-11 b\_2017-09-20.png (019f33b9e4d0...), FinalLightCurve\_WASP-11 b\_2017-09-20.csv (48164e69b694...)

## Compute resources

Wall time 85.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 10:35Z.